Fundamentals of Programming I

Chapter 2 - Basic Programming Constructors Practice Exercises

Part 1: Coding Writing

How to use this page

- Type each program in your IDE, compile it and run it. Do not just read it.
- Test it with the given data, then with your own.
- Only open the solutions page after you have tried.
- These use only what Chapter 2 teaches: input/output, variables, operators, the conditional operator, casting and the math library. Decisions with `if` and loops come in Chapter 3.

Easy level

Exercise 1: Escape sequences

Print exactly two lines. Line 1: the word Name:, a tab, then Abebe. Line 2: She said "Hello" (the quote marks must appear in the output).

Test data: Expected output: Name:[tab]Abebe and She said "Hello"

Exercise 2: Sum, difference and product

Read two integers. Print their sum, difference and product.

Test data: 7 and 3 give 10, 4 and 21.

Exercise 3: Rectangle area and perimeter

Read the length and width of a rectangle (decimals allowed). Print the area and the perimeter.

Test data: 5.5 and 2 give area 11 and perimeter 15.

Exercise 4: Simple interest

Read the principal P, the rate R (percent per year) and the time T (years). Print the simple interest $I = P \times R \times T / 100$.

Test data: P = 2000, R = 5, T = 3 gives 300.

Exercise 5: Average of three numbers

Read three integers and print their average as a real number.

Test data: 5, 6 and 8 give 6.33333 (not 6).

Medium level

Exercise 6: Seconds to hours, minutes, seconds

Read a number of seconds. Print it as hours, minutes and seconds.

Test data: 3725 gives 1 h 2 m 5 s.

Exercise 7: Largest of three (conditional operator)

Read three numbers and print the largest. Use only the conditional operator ?: (no if).

Test data: (4, 9, 6) gives 9. (7, 7, 3) gives 7.

Exercise 8: Swap two numbers

Read two numbers. Print them, swap their values using a third variable, then print them again.

Test data: 10 and 25: before 10 25, after 25 10.

Exercise 9: BMI with formatted output

Read weight (kg) and height (m). Compute BMI = weight / (height × height). Print a small table with the three values, two decimal places each, in columns of width 12.

Test data: 70 and 1.75 give BMI 22.86.

Exercise 10: Sum of the digits of a 3-digit number

Read a 3-digit integer and print the sum of its digits.

Test data: 482 gives 14.

Hard level

Exercise 11: Leap year

Read a year. Print "Leap year" or "Not leap year". A year is a leap year if it is divisible by 4 but not by 100, or it is divisible by 400. Use a logical expression and the conditional operator.

Test data: 2024: leap. 1900: not leap. 2000: leap. 2023: not leap.

Exercise 12: Reverse a 3-digit number

Read a 3-digit integer and print it reversed.

Test data: 123 gives 321. 405 gives 504. 120 gives 21 (explain why).

Exercise 13: Distance between two points

Read the coordinates (x1, y1) and (x2, y2). Print the distance $\sqrt{((x2-x1)^2 + (y2-y1)^2)}$ with 2 decimals. Use the math library.

Test data: (1, 2) and (4, 6) give 5.00.

Exercise 14: Wrapping around

Declare an `unsigned short` equal to 65535. Print it, add 1 and print again, then subtract 1 and print again. Explain the results.

Test data: Expected: 65535, then 0, then 65535.

Exercise 15: Bitwise operators

Read two integers a and b. Print a & b, a | b, a ^ b, ~a, a << 2 and a >> 1. Then use the conditional operator and the expression (a & 1) to print whether a is even or odd.

Test data: a = 8, b = 3 give 0, 11, 11, -9, 32, 4, and even.

Part 2: Operator Precedence, Worked by hand

No program needed. Use pen and paper, show your steps, and use the table below. Then check yourself on the solutions page.

Precedence table (highest first)

Level	Operators	Order
1	`++ --` (postfix)	left to right
2	`! ++ --` (prefix), `sizeof`	right to left
3	`* / %`	left to right
4	`+ -`	left to right
5	`< <= > >=`	left to right
6	`== !=`	left to right
7	`&&`	left to right
8	`||`	left to right
9	`? :`	right to left
10	`= += -= *= /= %=`	right to left
11	`,`	left to right

Rules to remember

- Brackets `( )` are done first.
- Operators on the same level follow the direction in the last column.
- Relational and logical results are **1** (true) or **0** (false). In arithmetic they behave as 1 and 0.
- Assignment operators (including `+=`, `-=`, `*=`, `/=`, `%=`) are the lowest. The whole right side is worked out first.
- `&&` and `||` stop early: if the left side already decides the answer, the right side is not evaluated, so its `++` or `--` never happens.
- Dividing two integers drops the fraction.

A. Add brackets (no calculation)

Question 1 Easy

```
n <= p && n >= p || n == 0
```

Fully bracket the expression to show the order of evaluation.

Question 2 Easy

```
a + b * c - d / e % f
```

Fully bracket the expression.

Question 3 Medium

```
x += y = z * 2
```

Fully bracket the expression.

Question 4 Medium

```
!a == b && c
```

Fully bracket the expression.

Question 5 Hard

```
a = b > c ? d : e + 1
```

Fully bracket the expression.

B. Work out the result by hand

Question 6 Easy

```
int r = 2 + 3 * 4;
```

Find r.

Question 7 Easy

```
int r = 20 - 6 / 2 + 1;
```

Find r.

Question 8 Easy

```
int r = 17 % 5 * 2;
```

Find r.

Question 9 Easy

```
int r = 10 / 4 * 2;
```

Find r.

Question 10 Easy

```
int r = 3 + 4 > 6;
```

Find r.

Question 11 Medium

```
int a = 5, b = 3;
a += b * 2;
```

Find a and b.

Question 12 Medium

```
int x = 10, y = 4;
x -= y + 2;
```

Find x and y.

Question 13 Medium

```
int a = 12, b = 5;
a %= b + 1;
```

Find a.

Question 14 Medium

```
int a = 3, b = 7, c = 5;
int r = a > b || c == 5 && b > a;
```

Find r.

Question 15 Medium

```
int x = 5;
int r = !x > 3;
```

Find r.

Question 16 Medium

```
int a = 5;
a += a > 3;
```

Find a.

Question 17 Hard

```
int x = 4, y = 9;
x += y > 5 && x < 10;
```

Find x.

Question 18 Hard

```
int a = 6, b = 4;
a -= b *= 2;
```

Find a and b.

Question 19 Hard

```
int a = 5, b = 2;
a *= b += 3;
```

Find a and b.

Question 20 Hard

```
int a = 5, b = 10, c = 15;
int r = a + b > c || !(c - b == a);
```

Find r.

Question 21 Hard

```
int x = 7;
x %= 4 + 1;
x *= x > 1;
```

Find the final x.

Question 22 Hard

```
int a = 1, b = 2, c = 3;
int r = a == 1 || b++ > 5 && c--;
```

Find r, b and c.

Question 23 Hard

```
int x = 5, y = 0;
int r = x > 3 && y++ < 1;
```

Find r and y.

Question 24 Hard

```
int x = 3, y = 5;
x += y;
y -= x;
x *= y;
```

Find the final x and y.

Question 25 Hard

```
int a = 7;
int r = a > 5 ? a *= 2 : a /= 2;
```

Find r and a.